Research proof for review

# Optimal fractional vertex cover under vertex arrivals

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In the vertex-arrival model, each new vertex reveals its edges to earlier vertices. A fractional algorithm maintains a value $y_v \in [0,1]$ on each vertex. Values can only increase, and every revealed edge $uv$ must satisfy $y_u + y_v \geq 1$. On an unweighted input, ALG is the final sum of these values and $\mathrm{OPT}_f$ is the minimum cost of an offline fractional cover.

**Theorem.** For every $\epsilon > 0$ and every randomized online fractional vertex-cover algorithm, there is a finite, unweighted bipartite vertex-arrival input, independent of the algorithm's realized random bits, on which

$$\mathbb{E}[\mathrm{ALG}] \geq (B^* - \epsilon)\,\mathrm{OPT}_f.$$

Here $k > 1$ solves $k = \coth k$, $a = (k+1)/2$, $b = (k-1)/2$, and

$$B^* = 1 + a^{a/k} b^{b/k} = 1.9007616968738465\ldots.$$

The core inputs are blowups of rooted trees. Each new pool is adjacent to just one old pool, and old pools can receive arbitrarily many children. Together with Wang and Wong's upper bound [1], this gives the optimal ratio also for the general-graph fractional problem.

The proof keeps track of the worst cost separately according to whether the offline cover includes the root. After a child arrives there are three compatible pairs of root choices. A full-history strategy might try to spend less by observing which of the two branches grows first. Sections 2 and 3 show that the adversary can order the two tests so this information has no value at either branch's decisions. Section 4 then extracts Wang and Wong's function from the frontier of those conditional costs.

## 1 A countable game and its conditional costs

Normalize the root pool to weight one. A core move adds an independent pool of announced positive rational weight joined completely to its named parent. There are no other new edges. Number pools in local arrival order and use the opposite color of the parent. Allow every finite such history, including a stop just after the root. An online response gives monotone coordinates in $[0,1]$ satisfying each edge inequality. A pool represents an average over twins. Section 5 removes the batch and rational-weight conventions.

Fix a rational $0 < \beta < B^* - 1$. For any integral cover $C$ of a core prefix, put

$$L(C) = \sum_{v \notin C} w_v y_v - \beta \sum_{v \in C} w_v.$$

A strict positive value gives a finite violation of $1 + \beta$. Attach a private leaf to each individual vertex represented by $C$. The completed graph has an offline cover $C$ and a disjoint private matching of the same cardinality. Monotonicity preserves the weight already paid

outside $C$. Each private matching edge requires at least one unit from its endpoints. Thus, after integer scaling, the algorithm pays at least $w(C) + \sum_{v \notin C} w_v y_v$, and $\mathrm{OPT}_f = w(C)$, where $w(C) = \sum_{v \in C} w_v$. Lu uses a complete-bipartite extension with the same matching accounting [2]. The edge-arrival construction of Tang and Zhang uses private edges [3].

Start the isolated root at zero. Define $F(p)$, for real $p \geq 0$, as the least $q$ for which a **single causal strategy on all core histories** can maintain both promises

$$L(C) \leq p \quad \text{(root outside } C\text{)}, \qquad L(C) \leq q - \beta \quad \text{(root inside } C\text{)}. \tag{1}$$

Assume for contradiction that all finite families of core prefixes admit a strategy with $L(C) \leq 0$ for every prefix and cover. The complete strategy space uses countably many coordinates in $[0, 1]$, one vector per public history. Feasibility, monotonicity and the promises at a given prefix are closed finite-coordinate conditions. Compactness, equivalently diagonal subsequence extraction, gives one strategy for all histories. Thus $F(0) \leq \beta$. The root-only inside test gives $F \geq 0$. The infimum is attained by compactness. The attainable epigraph of $(p, q)$ is closed and convex by compactness and coordinatewise mixing. Consequently $0 \leq F \leq \beta$, $F$ is nonincreasing and convex, and it is continuous on $[0, \infty)$, including at zero by closedness and monotonicity. No scalar state assumption has been made about the algorithm.

## 2 Interleaving two finite tests

A *local test* is a pair consisting of a finite rooted input and a terminal cover of that input. Only the sequence of core arrivals is public. The terminal cover is a hidden label. A finite distribution of tests is a tree of public prefixes, with termination and its hidden label as possible final outcomes. Several outcomes may have the same last public prefix.

**Interleaving lemma.** Take two finite rational distributions of tests on disjoint rooted components $A, B$. Write $\alpha = \Pr(A\text{'s root outside})$ and $\delta = \Pr(B\text{'s root inside})$. If $\alpha \leq \delta$, there is a distribution of interleaved tests with these properties.

1. Both local marginals are preserved, including their entire input and hidden cover. The two selected covers jointly cover the connecting root edge on every outcome of positive probability.

2. At **each local arrival** in either component, conditional on its local public prefix, the joint public past available for that decision gives no further information about the rest of that component's test. This includes its hidden cover.

**Construction.** In the first local prefix tree mark each node by the conditional probability that its terminal cover excludes its root. In the second tree use the probability that its cover includes its root. Terminal labels are marked zero or one. Each mark is the probability-weighted mean of its children's marks. Implement a multiway transition by a finite binary tree of choices, marking each internal split with its corresponding weighted mean. Only the resulting next public arrival is revealed. Zero-probability outcomes can be dropped. Choose a common denominator $M$ for all marks.

Place two particles at $M\alpha, M\delta$ in $\{0, \ldots, M\}$. They use the **same sequence** of independent fair signs. At each tick add the sign to each interior particle and leave an absorbed particle fixed. For a local binary choice with distinct child marks, wait until its particle first

hits either child mark on this lattice and take that branch. The gambler's ruin hitting probabilities give exactly the required two weights. For equal child marks use a fresh independent coin with the prescribed probability and advance no ticks. Continue this procedure through local arrivals and the final hidden label. Once absorbed, all remaining choices in that local tree have equal marks and use the coin rule.

Merge the local arrivals by tick, keeping local order. Break ties by a fixed component priority. Reveal only actual arriving pools and their parents. Hidden transitions and termination labels are not announcements. The particles remain ordered at every tick, including when one is absorbed. Every required hit occurs almost surely and there are only finitely many local transitions. At termination the two binary marks are ordered. Thus if $A$'s root is outside, $B$'s root is inside. Although the number of private walk ticks is unbounded, the public input ranges over only finitely many local paths and shuffles.

For the information assertion, the tick of a specified local arrival (on the event it is reached) is a stopping time of the signs and the coins used to reach it. At that time its particle is at exactly the mark of its **public prefix**. Future signs and fresh own coins give the same law for its remaining test as in standalone sampling from that prefix. The other component's arrivals already delivered are determined by the signs up through that tick and the other component's independent coins. This holds for ties as well, since their order does not depend on future signs or unused own coins. One may condition on all other-component coins and the stopped sign history in this argument. One must not condition on unused own coins. This proves assertion 2. At intermediate ticks, while a local move is pending, the other branch may convey information. No decision on the pending branch is required then.

## 3 Why the budgets must separate

Fix $p \geq 0$ and a strategy attaining (1) at $q = F(p)$. Offer a first child of rational weight $h > 0$. Fix the resulting root coordinates $r, z$, with $r + z \geq 1$. There are now two rooted components if their connecting edge is set aside. Each can undergo every rational rooted-tree continuation, in any interleaving. Scale the child's subsequent weights by $h$. Its local surplus will be expressed in units of $h$. Arrival indices in each local test can be translated canonically to global indices.

We claim there exist **two independent local strategies**, starting at $r, z$, that meet the global promises against *every compatible pair* of local tests. This is a consequence of the interleaving lemma, even though the original strategy sees both histories.

If there were no such pair of strategies, compactness would give finitely many incompatible performance inequalities. For each test pair the inequality is that $L_A + hL_B$ be at most $p$ or $q - \beta$, according to $A$'s root status. Compatibility here means the two roots are not both outside. The product of local strategy polytopes on these finite prefix trees is compact and convex, and each performance inequality is affine. Finite minimax, or separation from the nonpositive orthant, gives a distribution on these compatible pairs for which **every independent strategy pair** has strictly positive expected excess above the appropriate threshold. Its weights can be taken rational by approximating within the strict margin, uniformly on the bounded strategy polytope.

Keep only the two local marginals of this distribution. Their ordered root probabilities allow the interleaving construction in Section 2. The expected threshold depends only on $A$'s

marginal root status. The expected performance of independent strategies likewise depends only on the two marginals.

To apply this argument to the original joint strategy, first postpone gratuitous raises. Simulate it internally, copy each newborn coordinate, and raise only its neighbors as needed for that coordinate. Induction shows that the resulting feasible coordinates are bounded above by the simulated ones at every prefix. This applies after the connecting edge with $r, z$ fixed. In particular the strategy changes coordinates in a component only at its own arrivals. It still meets (1).

Average this joint strategy's local vector at each local public arrival conditional on that local prefix. By the lemma the mean is unchanged if one also conditions on any particular subsequent local input and terminal cover of positive conditional probability. Thus means at consecutive local prefixes obey monotonicity, and all local edge inequalities hold by linearity. Before the first local continuation use the fixed initial coordinate. These are feasible independent local strategies on the sampled prefixes and can be extended off support by raising coordinates to one. For each terminal test the expected local surplus of the joint strategy is exactly that of its mean strategy. This remains true if the last arrival precedes other-component moves or if there is no local arrival. The combined expected excess must therefore be strictly positive. This contradicts the joint strategy's promises on every interleaving.

For the independent strategies let $s, t$ be the suprema of their outside-root surpluses. Both are finite by pairing with the other root's root-only inside test. They are at least $r, z$ by the root-only outside tests. A local strategy starting at a positive coordinate can be simulated from zero by postponing that initial purchase until the first local move. This can only improve its root-only outside test. By the definition of $F$, the supremal inside surpluses are at least $F(s) - \beta$ on the parent side and $F(t) - \beta$ on the child side. The two inputs can be chosen separately for any compatible pair of root statuses. Taking suprema in the outside/inside and inside/outside inequalities proves

$$s + h(F(t) - \beta) \le p, \qquad F(s) + ht \le F(p), \qquad s + t \ge 1. \tag{2}$$

This holds for **every** $p \ge 0$ and rational $h > 0$. Artificial initial promises $p$ need not be reachable from a particular play. Their frontier was defined using the very same rooted input family. At the isolated root we obtain $F(0) \le \beta$.

### 3.1 A fork that can be checked directly

As a check on the three compatible root choices, take two adjacent unit pools and allow one further child of weight $h = 1/2$ on each endpoint, in either order. The exact excess ratio with terminal completions is $3/5$. Label the endpoints $U, V$ so that $V$'s initial coordinate is at least $1/2$. Attach the first child to $U$, and let $r$ be $U$'s coordinate then. The cover consisting of $U$ requires $1/2 + h(1 - r) \le \beta$. Attach the second child to $V$ and write $s$ for $V$'s coordinate. The two covers that exclude one endpoint require

$$r + h(1 - s) \le \beta(1 + h), \qquad s + h(1 - r) \le \beta(1 + h).$$

Eliminating $s, r$ gives $\beta \ge (1 + h)/(2(1 + h^2)) = 3/5$. Conversely start both endpoints at $1/2$. If a child arrives, raise its parent to $4/5$ and set the child to $1/5$. Each local outside surplus is at most $1/2$, and each inside surplus is at most $-1/2$. Every compatible pair

therefore has nonpositive sum. The upper strategy uses only local histories, while the lower argument permits both histories to influence every response.

## 4 Recovering the exact constant

Put $D = \beta - F$. It is nonnegative, concave, and bounded by $\beta$. For a differentiability point $0 < p < 1$, convexity and (2) give

$$t \le -F'(p)D(t), \qquad t \ge 1 - p - h\beta.$$

Indeed $F(s) \ge F(p) + F'(p)(s-p)$, and $s - p \le hD(t)$. For any $\eta > 0$, $v/(D(v) + \eta)$ is nondecreasing for $v \ge 0$ by concavity and $D(0) \ge 0$. Let rational $h \downarrow 0$. Continuity on the interior yields

$$-F'(p) \;\ge\; \frac{1-p}{\beta + \eta - F(1-p)}.$$

Define $f_\eta(u) = \beta + \eta - F(1-u) > 0$ on the closed unit interval. Integrate from $p$ to 1, using local absolute continuity of convex functions, $F(1) \ge 0$, and continuity to extend to $p = 0$. Then

$$f_\eta(1-p) + \int_p^1 \frac{1-u}{f_\eta(u)}\,du \;\le\; \beta + \eta \qquad (0 \le p \le 1). \tag{3}$$

The function is continuous and $(1-u)/f_\eta(u)$ is nonincreasing by the same concavity argument. The infimum over this class of the supremum of the left side of (3) is $B^* - 1$. Letting $\eta \downarrow 0$ contradicts the choice of $\beta$. This invokes Wang and Wong's optimization theorem only after deriving its hypotheses from arbitrary causal cover strategies [1, Section 4].

## 5 Constructing a finite ordinary input

For a requested $\epsilon$, choose a rational $\max(0, B^* - 1 - \epsilon) < \beta < B^* - 1$. The following terminating search specifies finite adversarial families even when a manageable size bound is unavailable.

1. For $n = 1, 2, \ldots$, take all core histories with at most $n$ pools, root weight one, nonroot weights $a/b$ with $1 \le a, b \le n$, and every single-parent choice. Prefixes are shared across all continuations.

2. Test rational linear feasibility with one vector $y \in [0,1]^k$ per public prefix of $k$ pools. Impose monotonicity from its parent prefix, each new edge inequality, zero at the isolated root, and $L(C) \le 0$ for every integral pool cover at that prefix. Exact rational linear programming or Fourier–Motzkin elimination decides each stage. Stop at the first infeasible stage. If every stage were feasible, compactness would give the strategy ruled out by Sections 1–4. The weights used in the scaled child subgames are rational and thus all occur in this exhaustion.

3. Multiply all weights in this stage by $\operatorname{lcm}(1, \ldots, n)$. Replace each pool by that many independent twin vertices, presented consecutively, with all edges to the already present

parent pool. Against the given randomized algorithm evaluate expected coordinates on each fixed public prefix, averaging over twins at batch boundaries. These vectors satisfy feasibility and monotonicity. One may simulate it lazily with a zero initial root, obtaining vectors no larger than the actual ones, including if it makes gratuitous isolated purchases.

4. Infeasibility supplies a prefix and cover $C$ with strictly positive surplus on this evaluation. Finish that prefix with the private matching from Section 1. Since the lazy root-only surplus is nonpositive, the witness has a nonempty comparator. The selected sequence depends on the algorithm and its marginal behavior on fixed histories, never its realized seed. Its expected cost is strictly greater than $(1+\beta)w(C)$, in the common integer scaling. If the horizon is known, choose a common upper bound on the vertex counts of all these completed inputs, announce that horizon before evaluating the algorithm, and pad every shorter completion with isolated vertices.

All core and terminal edges are revealed with a new endpoint. The continuum limit in Section 4 tests rational offers of arbitrarily small size under the contrary hypothesis. The output graph itself uses no limiting weights or rounded player responses. The finite search is an existence construction and has no practical near-target runtime guarantee.

The script `scripts/tree_witness.py` implements the construction for a given algorithm. Its function `find_witness(beta, marginals)` evaluates `marginals(core_prefix)` on each scaled family, checks the cover budgets, and returns the completed unweighted arrival list, an offline cover, its optimum and the guaranteed ratio. This permits skipping the LP search for a family hard for every strategy. It must succeed no later than the first infeasible stage when given exact marginals and a subcritical target.

## 6 A concrete finite certificate at 1.9005

The saved certificate in `experiments/fork-1801-2000.npz` describes a particular finite family without enumerating all its interleavings. The following recursion explains how to expand it and why it is a lower certificate. It also checks the direction of the inequalities in (2) without numerical differentiation or evaluating the Wang–Wong function.

A record with rational $a, b \geq 0$ denotes a finite rooted test family on which the promises (1) require $q + bp \geq a$. A root-only record with intercept zero is valid for every nonnegative slope. If a strategy starts the root with a positive coordinate, the same lower bound holds by simulation from zero. At a fork, join two records with lines $a_j - b_j s$ on the parent side and $a_i - b_i t$ on the child side. Offer the child of weight $h$, then allow all order-preserving shuffles of their core tests. Allow stopping, and a private completion on every cover. Include root-only stops in each component. The finite separation argument of Section 3 gives $s, t \geq 0,\ s + t \geq 1$ and

$$p \geq s + h(a_i - \beta - b_i t), \qquad q \geq a_j - b_j s + ht.$$

For a desired $b > b_j$ with $bb_i < 1$, choose $h = (b - b_j)/(1 - bb_i) > 0$. Multiply the first inequality by $b$ and add the second. The coefficients of $s, t$ are now both $b - b_j$. Thus the forked family satisfies

$$q + bp \ \geq \ a_j + (b - b_j) - bh(\beta - a_i). \tag{4}$$

Rounding the right side down preserves validity. An alternative operation increases the slope of any old line at unchanged intercept using $p \geq 0$. This uses the same family.

Take $M = 4000,\ S = 4 \cdot 10^{12},\ \beta = 1801/2000$. Slopes are $\ell/M$ and intercepts are integer multiples of $1/S$. Row zero uses only root-only records. Record $(k, \ell)$ uses parent $(k, \ell - 1)$. A stored child index $i \geq 0$ uses child $(k - 1, i)$ and $h = M/(M^2 - \ell i)$ in (4). A value of $-1$ just increases the slope. Column zero is always the trivial line. These instructions are acyclic. They specify actual relative pool weights, allowed interleavings, and the arrival graphs for the witness in compressed form. Subtree weights are multiplied by the child's weight at each use and can all be given a common integer denominator on expansion.

At record $(126, 4105)$ the verified intercept numerator is 3,602,076,706,847, strictly above $S\beta = 3{,}602{,}000{,}000{,}000$. Thus $p = 0, q = \beta$ is impossible on that single family. The verifier recomputes the integer inequalities from the recorded child choices using Python arbitrary-precision integers. The certificate checks the finite claim conditional on the interleaving lemma. Sections 1–5 prove the limit independently of this example.

From the repository root, run the following command to replay the certificate.

```
python scripts/fork_certificate.py --check experiments/fork-1801-2000.npz
```

Running `python scripts/fork_certificate.py` without arguments regenerates it. The script `scripts/interleaving_check.py` separately checks a finite ordered walk with exact fractions, including coincident arrival ticks and an equal-mark split. It verifies the full conditional test law at each sampled decision prefix and that incompatible terminal labels have mass zero. These finite checks supplement the proof of the general lemma.

## References

[1] Yajun Wang and Sam Chiu-wai Wong. *Online Vertex Cover and Matching: Beating the Greedy Algorithm*. arXiv:1305.1694 (2013).

[2] Tianhang Lu. *A New Lower Bound for Online Vertex Cover under Vertex Arrivals*. arXiv:2608.09210 (2026).

[3] Zhihao Gavin Tang and Yuhao Zhang. *A Tight Bound on Online Vertex Cover under Edge Arrivals*. arXiv:2608.04994 (2026).